

Advanced VLSI Technology: FinFET Technology

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Abstract : In Today's market, scaling is one of most important and needy feature required and it is the most challenging job as it is difficult to scale with ideal values of the parameters. The channel length is reduced from micrometers to nanometers, but scaling problem is still not removed. In order to conquer this effect a new transistor FinFET, which controls the SCEs. FinFET Technology is considered to be favourable technology in designing over bulk MOSFETs. In this paper, an overview if FinFET is given along with comparison of MOSFET in terms of their operation. Analyses of Short Gate and Independent Gate FinFET and implementation of the some of the logic gates using FinFET.

Index Terms - FinFET, Short Channel Effect (SCE), Logic Gates, SG FinFET, IG FinFET.

I. INTRODUCTION

FinFET is a type of multi-gate Metal Oxide Semiconductor Field Effect Transistor (MOSFET). It was first developed at the University Of Berkley, California by Chenming Hu and his colleagues. A multi-gate transistor is the integration of more than one gate into a transistor. In FinFET, a thin Silicon fin is wrapped over the conducting channel, forms the body. The structure of this FET resembles with fins. The thickness of the device decides the channel length of the device. The channel length of a MOSFET is said to be the distance between the sources and drain junctions. It is a non-planar, double gate transistor which based either on the Silicon Wafers or on Bulk Silicon-On-Insulator (SOI). In order to fabricate compact and fast devices, the channel length of the device should be reduced constantly. The following parameters related to MOSFET apexes the need for compact, smaller devices and how MOSFET is not the acceptable choice for fabrication. The shorter section of the gate electrode is known as the length (L) and the longer section is called the width (W).

As the channel length of a MOSFET reduces, the short-channel effects increase. The short-channel effects are attributed to two physical phenomena:

- The limitation imposed on electron drift characteristics in the channel.
- The modification of the threshold voltage due to the shortening channel length.

The short-channel effects mainly are of five types:

Velocity saturation: Velocity saturation reduces trans-conductance in the saturation mode. When a strong electric field is energized, carrier velocity shoots to maximum value and becomes a saturated value, known as saturation velocity. When this occurs, the state of the transistor is known as velocity saturation. Due to Optical Phonon Emission, scattering rate of energized electrons get increased, causing velocity saturation. Hence, due to this effect the transit time of carriers through the channel get increased.

Impact Ionization: It usually occurs due to the high velocity of electrons in presence of high longitudinal fields that can generate electron-hole (e-h) pairs by impact ionization. Impact ionization occurs due to the impact on silicon and ionizing of the electron-hole pairs.

Hot electron effect: Hot electron effect occurs when electrons or holes gain high kinetic energy due to the presence of high electric field. As electrons has more mobility than holes so hot electrons are more probable than hot holes. Hot carriers get injected or trapped in certain areas and cause undesirable device behaviour and degradation hence give rise to hot carrier effects.

Surface Scattering: As the gate electrode channel length get reduced, the longitudinal component of the electric field shoots up, and the surface mobility looks like field-dependent. Surface scattering are the collisions suffered by the electrons which are accelerated toward the interface. The surface scattering causes reduction of the mobility. The carrier transport in MOSFET is limits within the narrow inversion layer. The mobility of electrons becomes difficult to move parallel to the interface. Due to this the average surface mobility becomes half as that of the bulk mobility.

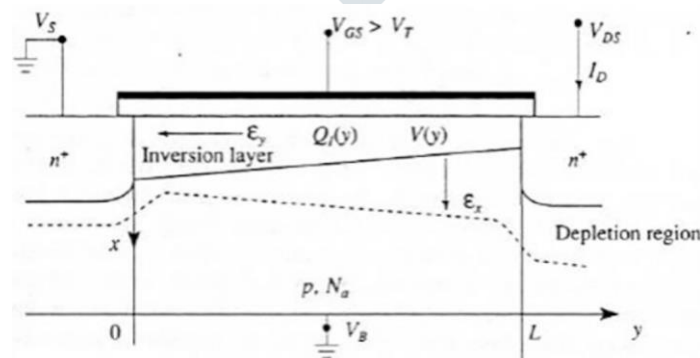


Fig.1 Surface Scattering

Drain-induced barrier lowering

The two depletion layer merge as a result of depletion region surrounding the drain which extends to the source. This leads to occurrence of punch through. Punch through can be reduced with the help of larger substrate doping, shallower junctions and thinner oxides.

Comparison of MOSFET and FinFET Operation:

MOSFET transistor: It consists of source, gate, drain, Silicon Germanium crystals, Nitride Spacer, Gate electrode, shallow trench isolation, the well, high-k dielectric and replacement gate metallization. Length 'L' and width 'W' are the short dimensions of the gate electrode and long dimensions of gate electrode respectively. When a gate electrode is energized with some voltage an electric field is induced which inverts the channel and forms an inductive passage between source and drain. When the length of the gate electrode is reduced by some distance then the control of gate over the channel also get reduced and this formation is drain induced barrier lowering, it increases source to drain leakage which hence lowers the transistor performance.

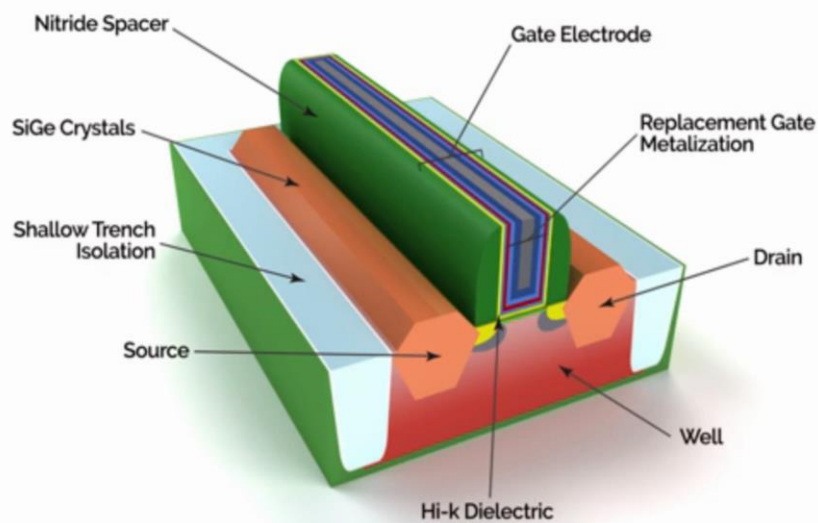


Fig.2 Internal Structure of FinFET

To overcome all these barriers in MOSFET, FinFET comes into account. In FinFET, gate electrode is wrapped over the channel. In this technology, a thin Silicon Fin acts as a channel and it is closed by the gate electrode on the three sides of the source and drain of those regions. The source drain regions consists of a silicon fin surrounded by an extension implant and polyoxide. The bottom interface oxide the high K dielectric and the Titanium Nitride work function metal the Tantalum Nitride, Titanium Aluminium, Tungsten and the additional layer of Titanium Nitride as part of the CMOS process.

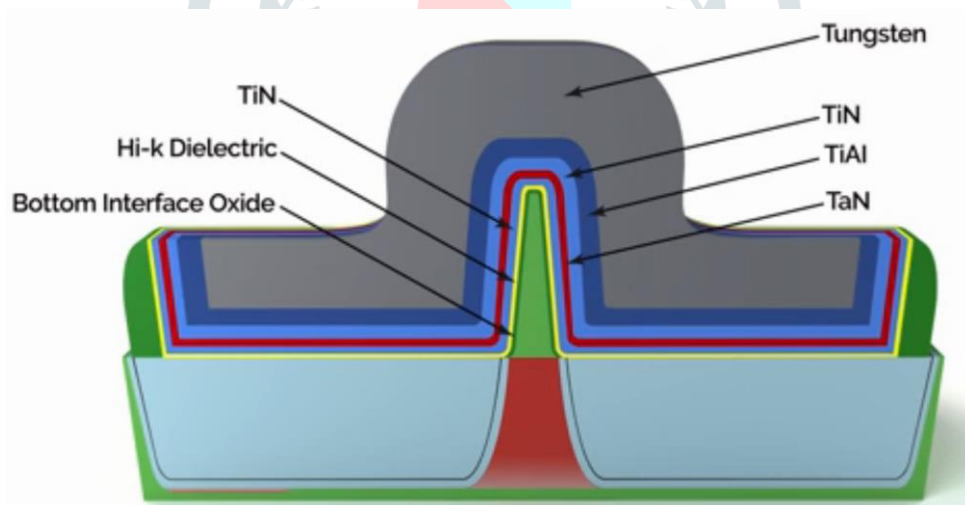


Fig.3 Different Layers of FinFET

When the gate electrode is energized, now the control of gate over the channel becomes more accurate as it surrounds the channel and the transistor, which consists of three components.

When the gate electrode is energized at the region of the fin then the gate electrode gets inverted and forms a conductive passage between the source and the drain as fin is the fully depleted device, most of the conduction occurs along the outer edges of the fin, the electrode is energized and transistor turns ON.

FinFET features Silicon Germanium crystals on the source and drain to strain the silicon fin and enhance the transistor performance. In order to implement the compressive strain, those portions of the fins which are outside the gate electrode must be removed which allows Silicon Germanium crystals to compress the channel region and enhance transistor performance.

CONCLUSION

FinFET is a promising substitute for bulk CMOS for meeting the challenges being posed by the scaling of conventional MOSFETs. Due to its double-gate structure, it offers innovative circuit design styles. Logic gates are implemented in SG-, LP-, IG-, and IG/LP-mode of FinFET. FinFET offer faster switching speed and reduces the leakage current. Logic gates net lists are simulated using HSpice simulator. From the simulation result propagation delay, power consumption, bandwidth, and power delay product can be obtained. The future work will address the implementation of Adder circuit and memory device using FinFET because of its high performance.

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FinFETs - Technology and Circuit Design Challenges

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Abstract - It took quarter of a century for multi-gate transistor to make it from first demonstration in research to a product – 22nm technology node microprocessor in 2012. FinFETs offer superior performance over incumbent planar devices due to their significantly improved electrostatics. FinFET technology faced two key barriers to their implementation in products: demanding process integration and its significant impact on layout and circuit design methodology. In this paper we focus on challenges and tradeoffs in both of these areas. Fin shape, pitch, isolation, doping, crystallographic orientation and stressing as well as device parasitics, performance and patterning approaches will be discussed. Implementation of high mobility materials for finFET devices will also be briefly reviewed as well as design challenges for logic and SRAM circuits.

I. INTRODUCTION

A. Why do we need finfets today?

Aggressive gate pitch scaling forced designs for 22nm Intel technology to use self-aligned contacts [1]. Beside contact, the width of two spacers and length of gate completely fills up remaining gate pitch distance (Fig. 1). Contact resistance grows exponentially with its decreasing width between two gate spacers, which in turn puts significant demand on scaling down gate length. This brings planar devices to the limits of their gate length scalability and necessitates use of alternative device architecture with improved electrostatic control of the channel.

Finfet device delivers much improved short channel control thru its fully depleted operation and requires low or no doping in the channel. Resulting lower transverse field in the channel improves subthreshold slope and allows for lower threshold voltages thus improving overdrive $V_g - V_t$ and consequently device performance. Lower channel doping also lowers dopant ion scattering leading to better drive currents and lowers random dopant fluctuations (RDF) [2], a critical factor in SRAM cells stability.

This paper focuses on issues introduced to technology and circuit design by introduction of finfets and does not address other challenges of scaling technology.

B. Brief history of finfets

The most significant era of multi-gate MOSFET discoveries happened about quarter century ago, in late 1980's. Likely the first multi-gate transistor was that published by Hieda et al. [3] in 1987. The authors realized that fully depleted body of this narrow tri-gate bulk Si-based transistor helps improve switching due to lessened body bias effect. Two years later, Hisamoto et al [4] demonstrated a predecessor of FinFET - first double gate transistor, in bulk silicon, called DELTA. First FinFET on SOI substrate was published a decade later [5]. SOI also enabled horizontal gate-all-around (GAA) transistor [6] creating a precursor to silicon nanowire devices. Stacking more than one nanowire on top of each other demonstrated increased drive current capability for a given foot-print size of a transistor [7].

The first demonstration of FinFET circuit was a 4-stage inverter by Rainey et al in 2002 [8] and the earliest report of FinFET ring oscillator was published by Nowak et al. [9]. FinFET SRAM cells have been reported in 2002 [10] and 20Mb SRAM array in 2004 [11].

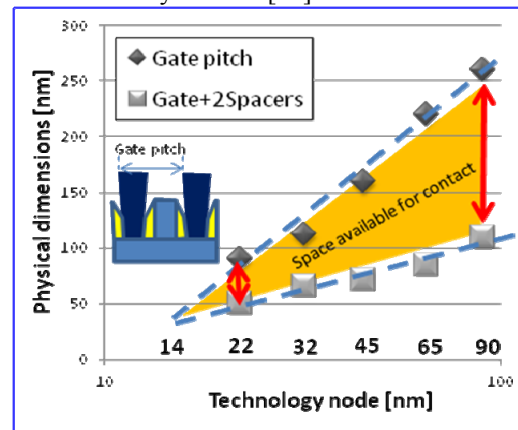


Figure 1 Gate pitch scalability challenge: Disappearing space for source/drain contact leads to increasing demand on gate length scaling.

II. TECHNOLOGY DESIGN CHALLENGES

Almost every aspect of device technology is affected by transitioning from planar to fin device architecture. We address the key issues below.

A. Fin patterning.

In order to match or exceed effective width of a finfet device with that of a planar device placed within the same footprint on the wafer, finfets need to be very tall or more of those need to be placed per pitch of active area. Generally, formation of two fins per minimum pitch allows reasonable fin aspect ratio that meets or exceeds effective width of corresponding planar device. Lithographic patterning of such fins has several key drawbacks:

- double patterning is required to halve the minimum pitch. Inherent overlay error between two fin patterns could lead to undesired fin pitch variation that impacts down-stream processing;
- desired fin width, roughly twice smaller than the smallest dimension that had been printed so far, that of gate length, is below well controllable capabilities of optical lithography, resulting in poor fin width control;
- line edge roughness (LER) of the process leads to substantial local fin width variability (LWR).

Using spacer-defined self-aligned double patterning (SADP) substantially alleviates dimensional problems and utilizes one mask layer rather than two. Utilizing 193nm immersion scanner allows in its limit to bring fin pitch down to ~40nm. Narrower fin pitches would require SADP method to be applied twice. Utility of SADP significantly lowers local variability of fin width caused by LER, as the spacer LER along its two edges largely mimic each other leading to much improved LWR, or fin width variation. EUV lithography, while capable of delivering better controlled line width for fins than optical lithography, may not be able to improve on the LER of the latter thus leaving SADP as the fin patterning method of choice for 7nm technology node and beyond.

Lithographic restrictions require highly regular patterns at 20nm node and below, hence, unidirectional fins on one pitch are highly desirable. Typical approach to fin patterning would be so called “sea-of-fins” where large areas of the chip are fully patterned with fins on one pitch. Unwanted fins and pieces of fins are subsequently removed by another litho-based patterning, so-called “fin-cut” masking step. Due to sub-lithographic fin pitch, removal of undesired single fins could be challenging for lithography and pairs of such sacrificial fins may need to be designated for removal. Careful designing is required to minimize chip area loss. Fig. 2 shows SEM image of fins and gates after gate etch.

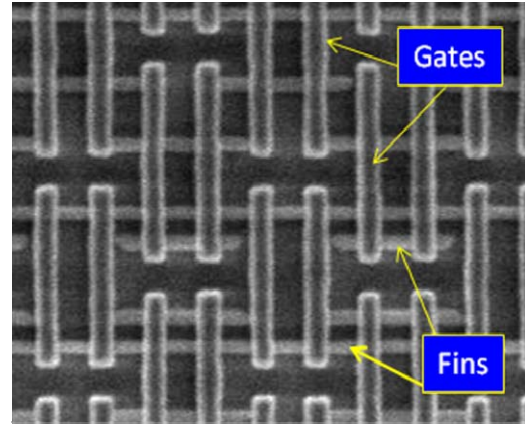


Figure 2 Top view SEM image of patterned gates over fins.

B. Fin shape

First finfet-based high performance logic product – Intel’s 22nm node microprocessor - has been built with finfet sidewalls sloping at about 8 degrees from vertical. Such shape has several practical reasons for manufacturability of this technology:

- Fins with lower aspect ratio (height:width) are more sturdy mechanically thus less vulnerable to damage during processing;
- Sloping sidewalls assure better fill of trenches between fins with fin isolation dielectric;
- Etching gate spacer off fin sidewalls (if desired) is easier than for vertical fins;
- Gate etch, which in finfets requires much more overetch, is easier;
- Doping of source/drain extensions by implantation is easier as sloped sidewalls are more suitable for dopant placement by vertical or low angle tilt implants. (In scaled technologies, tilt angle for implants of finfet body is usually limited to less than 10 degrees due to resist stack edge shadowing effect).

Sloping fin sidewall has a significant drawback – poor short channel control toward the bottom of the fin [12, 13] (Fig. 3). Such fins would usually require additional doping to alleviate this problem, thus sacrificing some drive current performance thru dopant ion scattering and causing increased random dopant fluctuation.

These negative aspects of sloping fin sidewalls will become more critical with scaling gate length and will need to migrate toward more vertical shape.

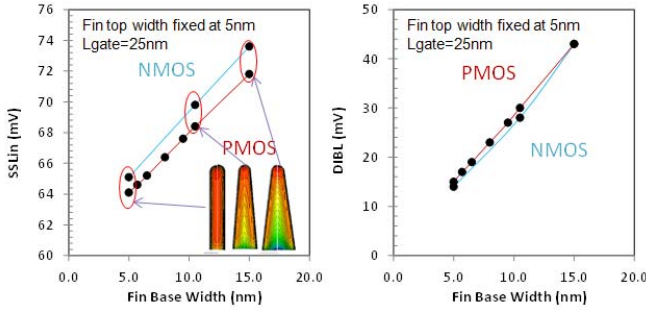


Figure 3 Modeling of short channel effects in finfets with different sloping of the sidewalls (varying fin base with fixed fin top width at 5nm) shows increasing deterioration of finfet device short channel integrity (DIBL and Subthreshold Slope) for finfets with more sloping sidewalls [13].

C. Fin dimension variability.

Fin height and width can have strong impact on device performance variability. Of the two, fin height variation is generally more critical [14]. Device's effective electrical width is directly related to fin height. Hence, any fin height variation due to variation of fin forming processes directly transfers to device width variation. Unlike in planar devices, where active area patterning variation affects only the narrowest of transistors, all fin-based devices suffer from the same percentage of device width error. This variability is primarily related to definition of fin height by STI dielectric recess process after polishing planarization and can reach several percent of every device width.

D. Device doping

Ideally, one would desire no doping in finfet channel. However, some light doping may be required to set alternative threshold voltages in certain devices or better control of under-the-fin leakage current. Those dopings are typically carried out with implantation. Source/drain doping, which requires high doses of dopant, faces significant challenge in terms of increased series resistance. This is related to implant damage in the fins that, due to fin's geometry, is retained in higher amounts than in corresponding planar structures [15, 16]. High temperature (300-400C) implants, plasma-based doping or so-called monolayer doping methods deliver dopants more conformally and with less damage to the fin. Alternatively, in-situ doped epitaxial material is deposited in source/drain area to deliver the dopant. This can be carried with or without removal of the fin in source/drain area prior to epi.

Selective epitaxy in source/drain area, besides providing dopants, can also deliver stress for improvement of channel mobility and shapes the contact area for subsequent silicidation. Two approaches can be pursued, with or without epitaxial growth on neighboring fins of the same device merging with each other. The former can provide more stress to the device by increased epitaxial stressor material volume if merging of the epi fronts does not result in defect generation that might relax the stress. The latter delivers more

area for placement of silicide contact and avoids excessive protrusion of the source/drain epi of the device's outer fins that can lead to shortening with the neighboring device.

E. Stress for fins

Fins can be stressed in a similar fashion to that of planar devices by selective epitaxy of higher or lower lattice parameter material than that of the channel material for desired PMOS or NMOS channel stress, respectively. Such stressing in fins is generally less effective than in planar devices and further diminishes with gate pitch and fin size scaling. Application of stress from under the fin channel by lattice-mismatched epitaxial buffer layer material (stress-relaxed buffer – SRB) has been found by modeling to be substantially more effective than source/drain stressors in scaled finfets [17]. Single SRB solution for NMOS and PMOS finfet is forecasted possible with SiGe SRB where Ge concentration is in mid range of about 40-50%. It provides tensile strain to Si fin for NMOS grown on top of it and compressive stress for SiGe with high concentration of Ge or a pure Ge fin grown for PMOS [18].

F. Fin orientation

Nominal orientation of fins, along $\langle 110 \rangle$ direction on (100) wafers results in finfet current flowing on (110) sidewall surfaces. Hole mobility is sizably higher on (110) surface than on (100) but the difference decreases with increasing strain [19]. Electrons flow somewhat slower along (110) plane than on (100) in planar devices. However, in finfets, quantum confinement results in quite different behavior – electron mobility becomes comparable or better for (110) sidewall conduction than for (100) [20].

Growth of epitaxial material on (100) fin surfaces results in uniform thickness increase that might be desired in some situations than that of diamond-shaped structures grown on (110) walls.

G. Fin isolation.

Source-to-drain leakage. Similarly to bulk planar devices, bulk finfet requires doped wells below the active part of the fin to assure prevention against source-to-drain punch thru leakage path. Such junction-based isolation will likely be insufficient for finfet devices with gate length $L_{gate} \sim 15\text{nm}$ thus demanding a more robust isolation below channel, in form of dielectric layer (Fig. 4). Such isolation can be achieved by utilization of SOI substrates or formation of local oxide under the bulk Si fin. A semiconductor buffer layer under finfet channel with appropriately engineered band structure could eliminate a need for junction or dielectric isolation in finfets [18].

Device-to-device leakage. Junction area between source/drain and substrate is much smaller in finfets than in planar devices. Consequently, leakage to substrate is lower and device-to-device isolation between finfets requires shallower trenches. In recent planar technology nodes STI trench depth was around 200nm. Finfets would require less than 100nm.

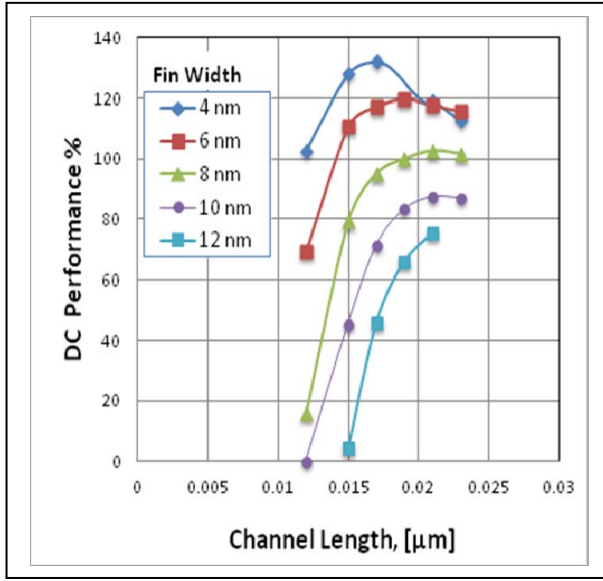


Figure 4 Simulated normalized drive current dependence on gate length and fin width in bulk Si finfet device. I_{off} set at 100nA/μm. Anti-punch-thru doping below fin and gate work function is set to satisfy off-current at constant level of 300pA/μm. Results indicate that device isolation with junction may not be sufficient for devices with ~15nm gate length, likely representing 7nm technology node.

H. Finfet parasitic capacitance vs planar.

Finfet has inherently higher parasitic capacitance than corresponding planar device. It consists primarily of gate-to-fin capacitance between part of the gate above the fin and the top of the fin, and can be optimized down to about 5% above planar device's. This capacitance decreases with decreasing fin pitch and increasing fin height, per effective device width [21, 22].

Bulk finfet junction capacitance between source/drain area and device well/substrate could be several times smaller than in planar devices.

I. Reliability

Finfet fully depleted operation provides lower transverse field in the device. This is credited for improved NMOS reliability for dielectric breakdown (TDDB) as well for threshold voltage instability (PBTI) observed in transition from planar 32nm to finfet-based 22nm technology node. PMOS reliability for both TDDB and NBTI appears unchanged for finfets [23].

J. Alternative fin materials

Scaling of finfet for 14, 10 and 7nm generations might be able to provide adequate short channel control but control of increasing power density will require novel solutions. Key element that has been utilized for such purpose over several generations of planar devices has been increased channel carrier mobility thru application of strain to channel material.

As silicon stressing will reach limits of its channel mobility improvements, an alternative channel material, with higher mobility, will be desired.

Leading contenders for such solutions are two different materials: III-V material is favored for NMOS, particularly InGaAs, and Ge or SiGe with high Ge content for PMOS. One type of material would be preferred for both n- and p-type devices, however, most of III-V materials exhibit poor hole mobility and Ge-based solution, while demonstrating high electron mobility, faces challenges related to germanium's high source/drain resistance. Recent work with ternary antimonides suggests that there might be a compromise solution available for both NMOS and PMOS in one channel material [24]. Silicon substrate will remain the material of choice for foreseeable future.

Implementation of two different channel materials on silicon substrate has its obvious challenges of cost and manufacturability.

Generally, three alternative integration schemes offer possible solution for CMOS fin devices (Fig. 5):

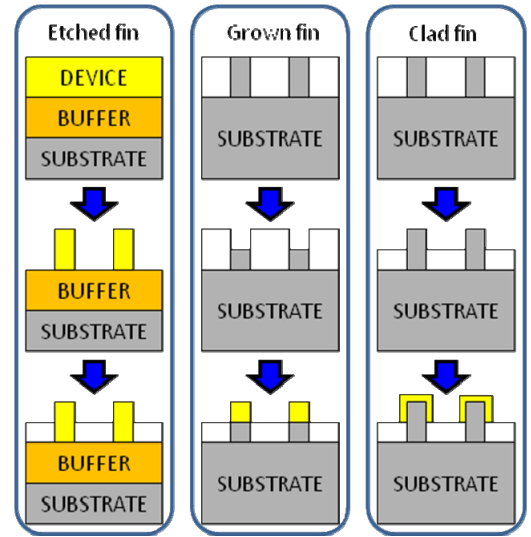


Figure 5 Fin formation with non-Si material on Si substrate. Fin material is grown epitaxially in blanket form, in fin-wide trench and around existing fin for the three schemes respectively.

- Etched fins. Large area epitaxial growth of two desired stacks of materials, comprising lattice mismatch accommodation and quantum confinement for channel material placed on top of the stack. Key challenges of such approach are: lowering sizable density of growth and stress relaxation related defects, and side-by-side accommodation of two thick and different materials.
- Grown fins. Epitaxial growth of individual fins in so-called "replacement fin" approach, where dummy silicon fins are entombed in STI dielectric, removed by etching and replaced by epitaxial growth of

desired fin materials from the bottom of fin trench. Much thinner epi films are grown in this approach. Defects related to lattice mismatch between substrate and buffer layer (or channel material, if buffer is not used) get largely trapped by trench walls and don't propagate to active part of the fin. Key challenge is to minimize lattice mismatch relaxation defects and avoid growth defects in very narrow trenches (<10nm) required for scaled fins.

- **Clad fins.** Cladding of structural fin (after STI formation and recess) with selective epitaxial growth of desired active fin material(s). With proper structural fin material choice, quantum confinement in such channel material is provided by adequate band structure of the fin material and current in such device would be largely confined to the cladding layer. For example, for SiGe PMOS cladding channel, silicon structural fin could be used, and for InGaAs NMOS an InP or InAlAs structural fin can be build using one of the two methods discussed above. Key challenge with this approach is the epitaxial growth of thin and uniform cladding film and the space constrains of narrow fin pitch.

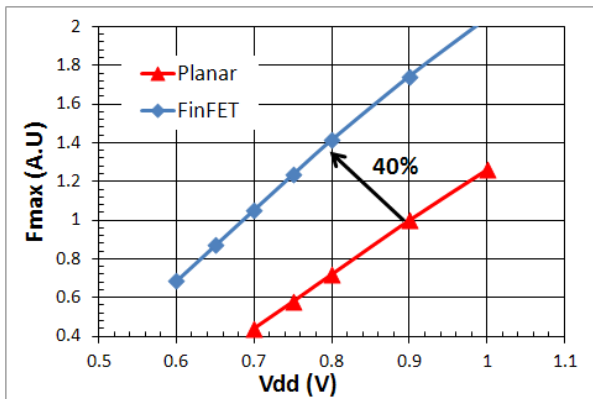


Figure 6 Calculated AC FOM for finfet and planar device technology indicates significant benefit of finfets at the same design ground rules.

K. Passive elements

Passive elements such as reference diodes and ESD can be realized either in long gated fin diodes and long channel finfets, or in the Si bulk substrate. Some resistors can be done in thin films (gate, MOL or BEOL metals), others in fins. Decoupling capacitors can be realized in fins or MIM capacitors.

L. Performance – planar vs. fin

Comparison of AC performance using compact modeling of figure-of-merit involving various logic gate configurations

indicates that as much as 40% gain can be realized in transition from planar to finfet technology at the same baseline design rules (itches) even with Vdd lowered from 0.9 to 0.8V (Fig.6).

III. LAYOUT DESIGN CHALLENGES

Converting planar device layouts to fins, faces the challenge of defining transistor widths in digital fashion forced by the one-size fins. Generally, wide devices that can be realized with large number of fins bringing about designed effective width will face little redesign challenge. Narrow devices and particularly SRAM transistors will face major redesign. STI width between SRAM n and p devices may require customized fin pitches, different than those used in the logic cells in order to further minimize SRAM cell size.

Generally all finfets on the wafer will have the same height for simplicity of processing, however, the three-dimensional character of transport in finfet offers another dimension in designing finfet devices. Taller fins can deliver more effective device width than planar ones (with height limits defined by process manufacturability). This design knob can either deliver better layout density than planar device or increase amount of drive current per given foot print. And as we mentioned earlier an added benefit of reduced parasitic capacitance is realized with taller fins.

Selecting the right fin pitch is not only related to process challenges but also requires optimization of “gear-ratio” between fin pitch and metal 2 pitch in standard logic cell design. But narrower fin pitch leads to more difficult “down-stream” processing such as trench filling for isolation, gate stack deposition and etch, etc. Careful optimization of those aspects is required in the design selection of the appropriate fin pitch.

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IV. SUMMARY

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Finfet technology has entered the market. High performance logic has adapted this device and will continue to use it for several generations into the future. New materials for fins are likely to be introduced into products in this decade. Substantial changes are brought up into circuit design world by finfet and design ecosystem is rapidly maturing with tools updates.

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Recent Trend of FinFET Devices and its Challenges: A Review

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Abstract— Recent technological demand of FinFETs have been explored and reviewed in this work. The downscaling of the conventional MOSFET urge to the researchers to innovate new electronic switches with improved performance in low power parameters and reduced short channel effects (SCE). Control of gate over the channel charge could be increased by using FinFET based multi-gate technology. In this review paper, we have studied about the parasitic capacitances and low power FinFET applications. The challenges of the FinFET technologies have also addressed to explore the research discontinuity.

Keywords—Short Channel Effect, FinFETs, Trigate, Parasitic Capacitance.

I. INTRODUCTION

Since the semiconductor technology is advancing, the electronic device structures are getting down-scaled day by day. The consequences of the device scaling reduce the controllability of gate over the channel charge and disturbed the flow of the charge through the channel causes the short channel effects (SCE) [1]-[3]. The impact of SCE deteriorates the device performance and makes the device unsuitable to scale down below 20nm. In order to increase the gate controllability over the channel charge, the semiconductor technology moves towards the multi-gate MOSFETs.

Multi-gate FETs are an alternative to planar MOSFETs, which improved the drain potential screening from the channel due to presence of additional gates [4]-[6]. In all multi-gate devices the two gate FETs (DGFETs) or tri-gate FETs are more desirable due to smaller parasitic capacitances and robust behavior against random dopant behavior. The Trigate FETs are having reduced fringing capacitances but at the cost of complex fabrication process [7]. The FinFETs are the emerging devices in this technological era which are having minimal power consumption, immune to SCEs, smaller area requirement and higher speed of operation [8]-[9]. Many of the available literatures reported about the symmetric and asymmetric FinFET structures and their implementation in digital and analog circuits [10]-[14]. However, all design hierarchy is not covered by FinFETs and also many circuit level implementations are untouched yet. The objective of this review paper is to provide a better understanding about the recent trend of FinFET technology and its applications. This paper comprises following sections: Section II states the history and its classifications, Section III deals with the challenging parameters in manufacturing process. The circuit domain issues have been explored in Section IV and lastly Section V summarizes review paper.

II. HISTORY AND CLASSIFICATION OF FINFETs

A. History of FinFETs

The fully depleted double gate SOI structure was first experimentally proposed by Hisamoto et. al. in 1989 [15], the first reported FinFET structure. FinFETs are having vertical channel known as FIN whereas in planar MOSFET horizontal channel is fabricated as displayed in Fig.1. Therefore, the channel width of FinFET is defined in terms of FIN height, which is known as width quantization. Hence, the flow of charge in the FinFET can be improved by increasing the number of fin on the structure which provides better gate control on the channel charge. The fin height is a measurable parameter to define the stability of the structure i.e. small fin height lead to a flexible structure compare to long fin structure [17]-[18].

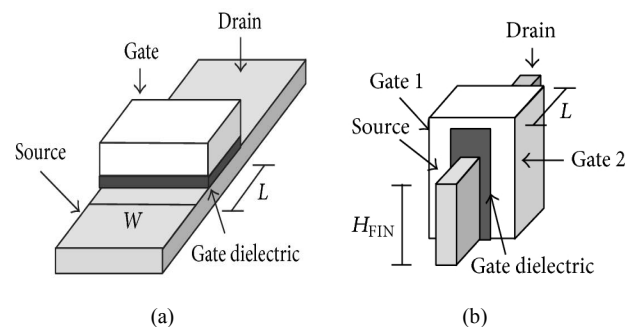


Fig. 1: Comparison between (a) Planar MOSFET and (b) FinFET.

FinFETs can also be implemented on the bulk wafers in which all fins share a common substrate. The bulk finFETs are easy to migrate from the bulk MOSFETs. In the planar SOI FinFETs, the fins are isolated. The comparison between bulk and SOI FinFET is shown in Fig. 2. The first FinFET based 4-stage inverter was reported by Guillorn et. al. in 2002 [23]. The first ring oscillator was proposed in 2008 by Nowak et. al. [19].

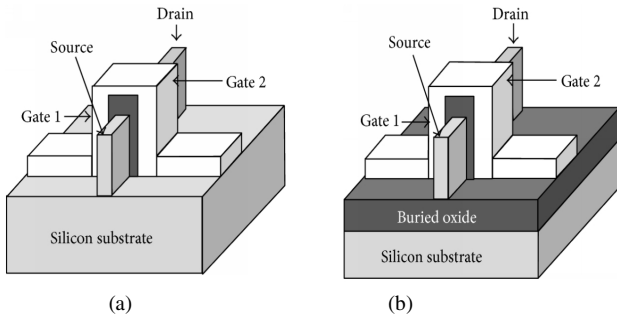


Fig. 2: Comparison between (a) bulk and (b) SOI FinFET.

B. Classification of FinFET Structures

The FinFETs are classified in basic two categories: (a) Independent gate FinFET (IGF) (b) Short gate FinFET (SGF). IG FinFETs are having four terminals while SGF are also known as three terminal FinFETs. The gates are isolated in IGF structures while the front and the back gates are shorted to each other in SGF structures as shown in Fig. 4. SGF structures are having higher ON current as compared to IGF structure because SGF structure jointly uses both the gates for electrostatic control of the channel. However, an IGF structure offers the flexibility to apply different signal on the different gates but at the cost of larger chip area.

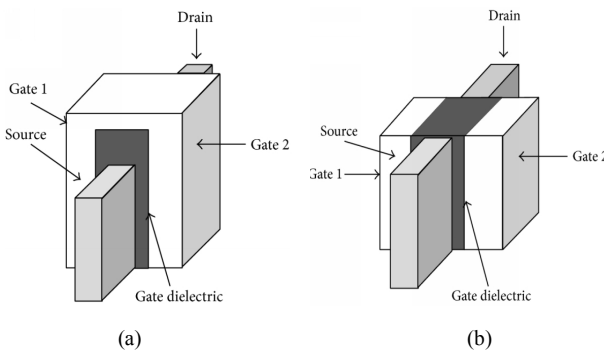


Fig. 3: Comparison between (a) SGF and (b) IGF.

On the basis of asymmetries the SGF structures can be further classified. We can keep the same work function of both the gate or we can take different value of work functions, known as Asymmetric SGF (ASGF) [20]. ASGF structures are very promising for short channel behavior [21].

III. CHALLENGING PARAMETER IN MANUFACTURING PROCESS OF FINFETS

The transition from the planar technology to the FinFET technology causes many challenges in manufacturing process. Few of them are addressed below:

A. Shape of Fin

The high performance logic function based on FinFET is an Intel microprocessor of 22 nm node technology. The fin is inclined by approx. 8 degrees from the vertical to built the microprocessor. The basic reason for this is the robustness of the FinFET structure, i.e. the FinFET having smaller aspect ratio (width-height ratio) is more robust. This inclination in the fin height causes the lack of immunity for short channel effects [22]. The sloping of the fin causes serious issues on gate length scaling.

B. Parasitic Capacitances

The shortcomings of the FinFET structures are having more parasitic capacitances as compared to planar structures. The additional increased overlap area across the front gate and back gate increases the capacitances of the device. The variation in the fin height helps to reduce the capacitance of the device, i.e. the increase in the fin height and decrease in the fin pitch helps to reduce parasitic capacitance of FinFET [23]-[24]. Analytical modelling of parasitic capacitances can be done by using conformal mapping technique [25]-[29].

C. Doping Concentration of FinFETs

The undoped channel is preferably for the FinFET technology [25]. However, to get better control over the leakage current the light doping is suggested. The doping concentration of Source/Drain region requires high dopant and hence increases the series resistance of the device. This damages the fin's geometry of the device. To overcome this problem in-situ epitaxial growth is done at the source and drain area with or without removing the fin of the structure.

D. Orientation of Fins

Generally, the flow of current on $\langle 110 \rangle$ sidewall surface is the result of fin orientation in $\langle 100 \rangle$ direction. The flow of electron is slow down in the direction $\langle 110 \rangle$ compare to $\langle 100 \rangle$ direction, while in FinFETs the opposite case happens. The quantum confinement effect causes better electron mobility in $\langle 110 \rangle$ direction [25].

E. Reliability of FinFETs

The lower transverse field in the device can be achieved by fully depleted FinFET operation. This improves the reliability of the NMOS for dielectric breakdown. The instability of the threshold voltage is also observed in transition from 32 nm planar technology to 22 nm FinFET technologies. The reliability for the PMOS appears unchanged in FinFET devices [30].

F. Variability of Fin Dimension

The fin height is an important parameter in FinFET technology as the electrical width of the device defined by the

fin height. In planar structures, the patterning of the device area affects the narrowest of the transistors, whereas the transition in width directly transferred to fin height. Therefore, fin height is a critical parameter [24].

G. Integration of the FinFETs

The advantage of compatibility with conventional CMOS processes, the FIN patterning and the gate formation step implored tighter control on fabrication steps as compared to planar technology[8]. For $L_G < 20\text{nm}$, uniformity of about 1nm for Fin width $< 10\text{nm}$ is necessary to maintain low threshold voltage variations. Hence, strict specifications for process variations need to be followed. Another critical issue is Gate patterning process step challenges due to the stack of the HF-based dielectric and the TiN/TaN/W/La₂O₃/Dy₂O₃/Al₂O₃ between metal gates are deposited to form the FinFET's complex gate stack and metal gate structures. Next to that, though basic body isolation between adjacent Fins are made using STI oxide, but still Fin channels are coupled underneath the oxide. This results sub-Fin Leakage [31]. This leakage is controlled by self-aligned punch-through stop layer (PTSL) at the bottom of the Fin.

H. Strain Engineering

As the Effective gate Oxide Thickness (EOT) of gate is reduced, constraint SiON gate dielectric scaling. This limit is further extended from strain engineering to increase the device performance insusceptible off-state behavior and short channel control [32]. Other options to improve the on-current may be uniaxial strain in the channel region using Source/drain stressors [33] or strained contact etch stop layers [34]

Self heating effect phenomenon, increase flicker noise, charge trapping in High-K dielectrics, reduced matching behavior are serious issues of analog circuit design. Few challenging aspects of circuit designs are discussed below:

A. Inverter VTC

When we analyze the FinFET inverter for the voltage transfer characteristics (VTC) and compare it with the CMOS based inverter in the near threshold region, the slope of VTC is steep in FinFET as compare to the existing CMOS technology. Crupi et al [35] compare the VTC curve for both FinFET technology and the CMOS based inverter for equal

drive strength of p-FET and n-FET and curve clearly shows the superiority of FinFET based circuits.

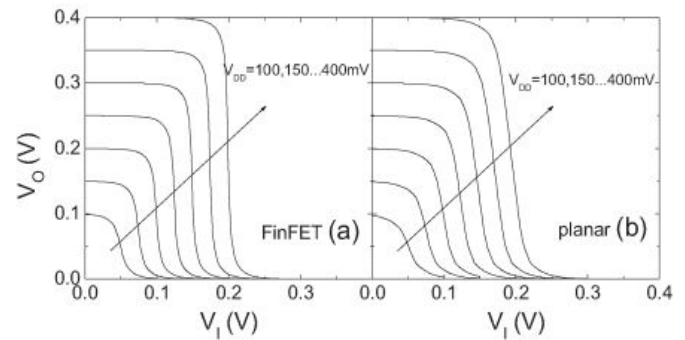


Fig. 4: VTC of FinFETs and Planar MOSFETs [25].

The drive strength for FinFET structures is matched but when we move towards complex circuit i.e. NAND and NOR gate, the matching of drive strength is very difficult due to the discrete device width measurement in terms of number of fins. When we design the FinFET based circuits the extension region depends on the transition time and decrease with the fan out as reported by Pandey et al[36]. Therefore, for designing the FinFET based circuits we have to look after demand of low active and leakage power dissipation reduction challenges.

V. CONCLUSION

The recent trend of FinFET technology is reviewed in this paper. The comparative study of the FinFET structure with bulk and SOI MOSFET has done with exploring different FinFET structures. In order to get better insight about the challenges of the FinFET technology, the circuit issues and manufacturing issues are addressed in the paper.

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